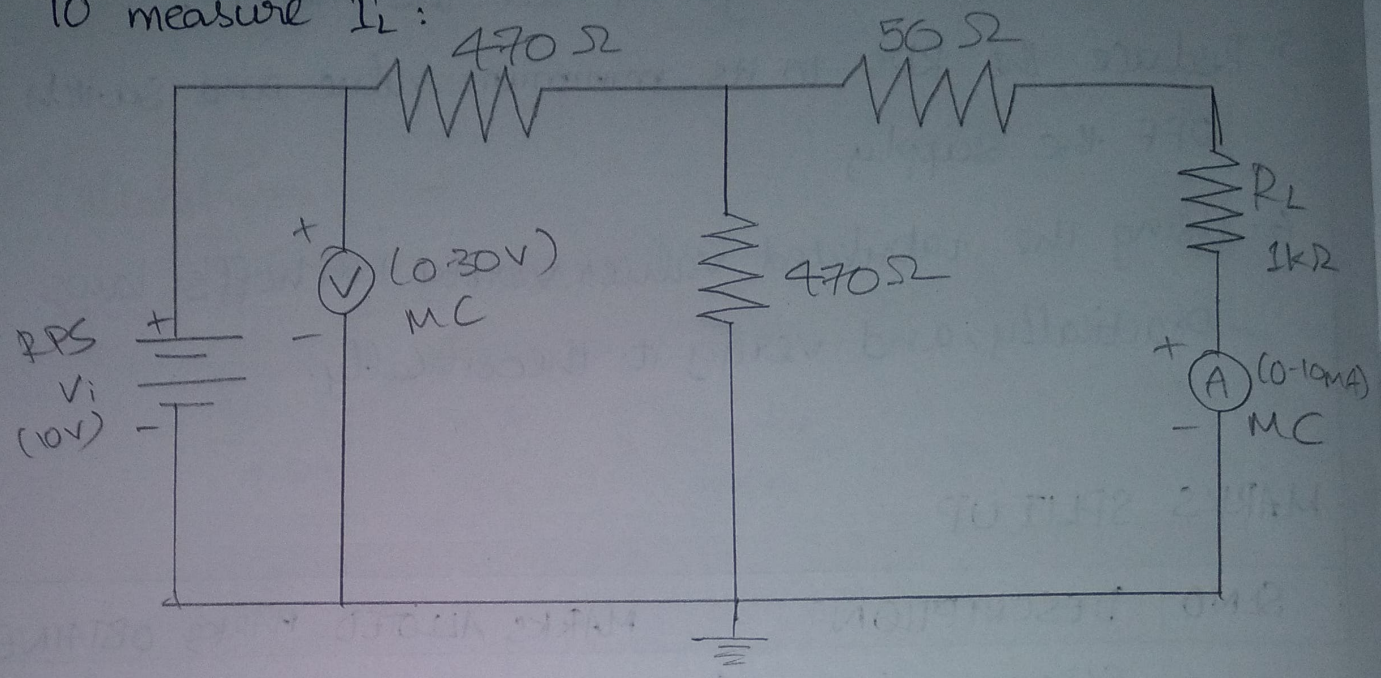
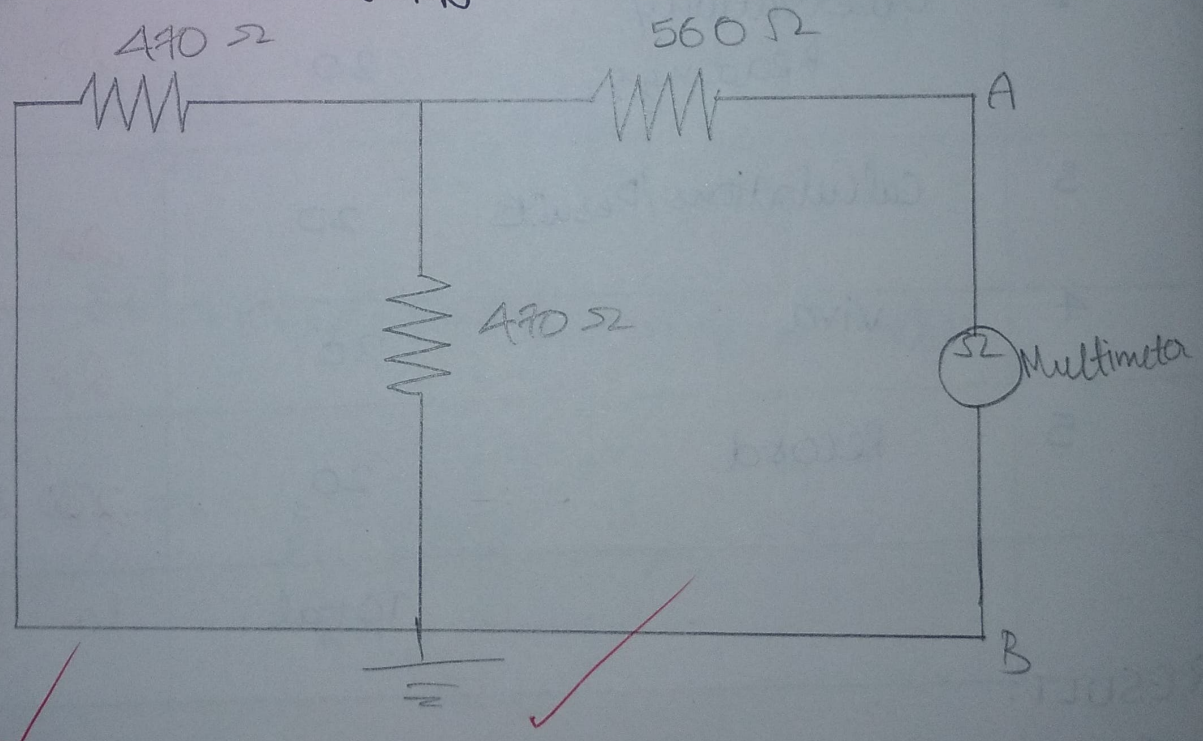
To measure I_L :



To measure R_{Th} or R_N



EXP. NO. 4 VERIFICATION OF THEVENIN'S THEOREM

DATE : 10.02.26

AIM :

To verify Thevenin's theorem practically and theoretically for the given DC circuit

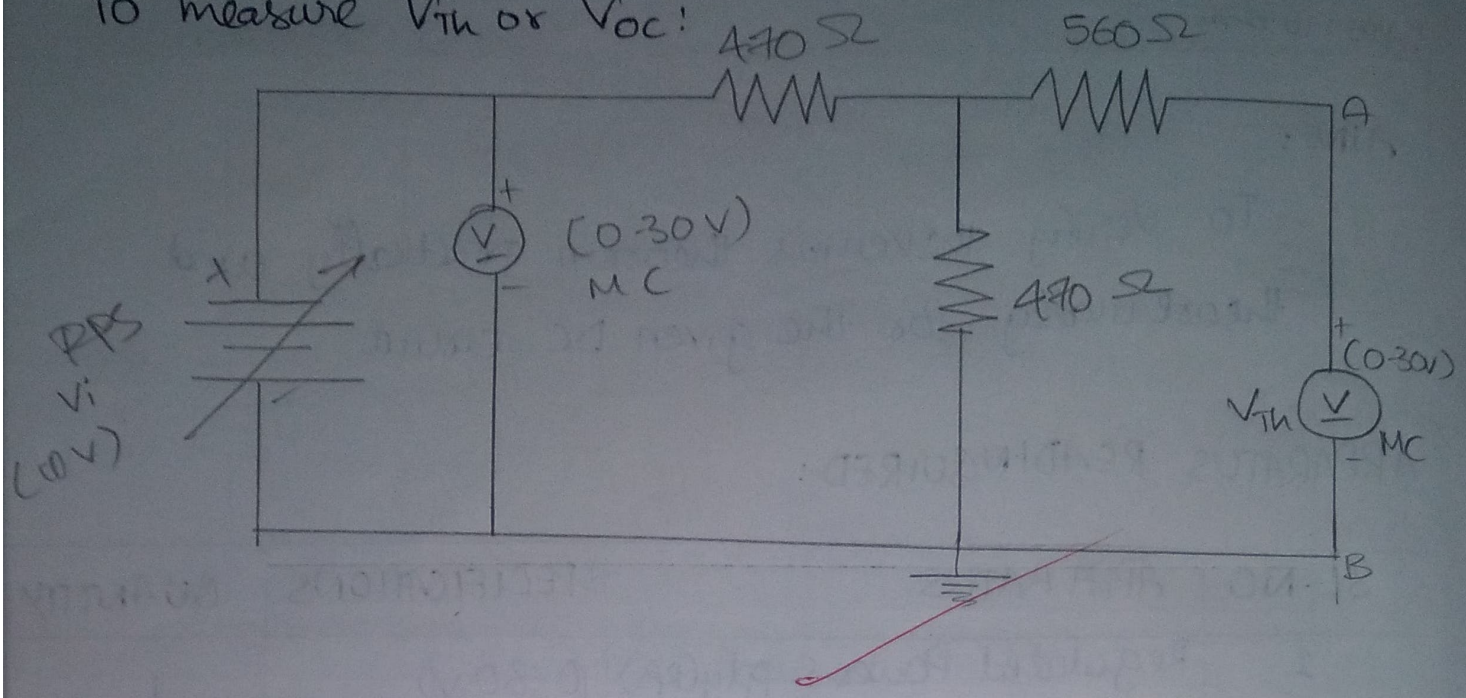
APPARATUS READINQUIRED :

SL.NO	APPARATUS	SPECIFICATIONS	QUANTITY
1	Regulated Power Supply (RPS)	(0-30 V)	1
2	Voltmeter	(0-30 V) MC	1
3	Ammeter	(0-10MA) MC	1
4	Resistors	470 Ω	2
		560 Ω	1
		1 kΩ	1
5	Bread Board	–	1
6	Multimeter	–	1

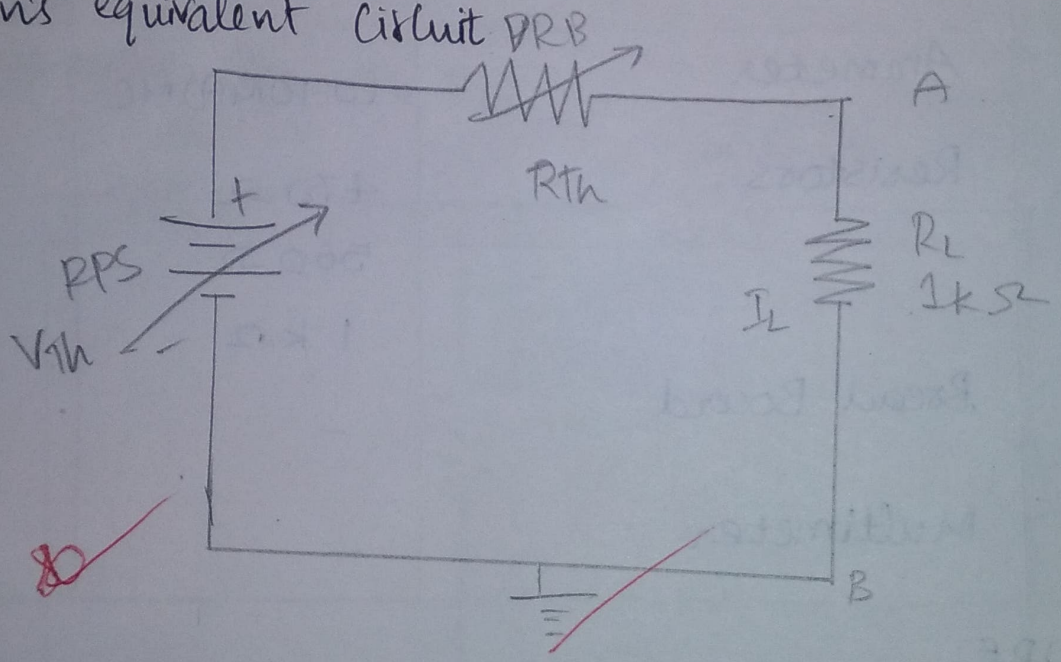
PROCEDURE :

1. Connect the circuit as per circuit Diagram 1 and set Vi = 10 V using RPS
2. Note Vi and IL readings in Tabular Column 1
3. Switch OFF the Supply and connect circuit Diagram 2 to measure RTh (=RN).
4. Switch OFF the Supply and connect circuit Diagram 3. Set Vi = 10V and note Vi and

To measure V_{Th} or V_{oc}:



Thevenin's equivalent circuit



To measure I_L

V_i	I_L
2 V	0.36 mA
4 V	0.72 mA
6 V	1.08 mA
8 V	1.44 mA
10 V	1.80 mA

O/P verified

VTh (= Voc) in Tabular column 3.

5. Switch OFF the supply and connect circuit Diagram 4. Set Vi = 10 V and note Vi and ~~VTh~~ IN (= Isc) in Tabular column 4.

6. Draw Thevenin's and Norton's equivalent circuits (5 & 6)

7. Calculate IL using

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

8. Verify Norton's Theorem by comparing theoretical and measured values

MARKS SPLIT-UP:

SI. NO.	DESCRIPTION	MARK ALLOTED	MARK OBTAINED
1	Experimental setup / connections	20	20
2	observation/Readings	20	20
3	Calculations/Results	20	20
4	Viva	20	20
5	Record	20	20
		Total	100

To measure R_{Th} or R_N:

from diagram 2 $\Rightarrow R_{Th} = R_N = 795\ \Omega$

To measure V_{Th} / V_{oc}

V_i	V_{Th}
2V	0.65V
4V	1.30V
6V	1.95V
8V	2.60V
10V	3.25V

$V_{Th} = 0.325 \times V_i$

10/2/26
0/8 verified

Model Calculation:

Given

Supply voltage = 10V

$R_1 = 470\Omega$ $R_3 = 560\Omega$

$R_2 = 470\Omega$ $R_L = 1k\Omega$

For calculating V_{Th} $\Rightarrow$

$$R_P = \frac{470 \times 470}{470 + 470} = 235\Omega$$

$$V_{Th} = V_i \times \frac{R_P}{R_1 + R_P} = 3.33V$$

(Practical = 3.25V)

For calculating R_{Th} $\Rightarrow$

470 parallel to 470

$= 235\Omega$

$R_{Th} = 560 + 235$

$= 795\Omega$

For calculating I_L $\Rightarrow$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$= \frac{3.33}{795 + 1000} = 1.85mA$$

10

11/2/26

Result:

Thus Thevenin's theorem is verified practically and Theoretically.